

Personal Library – End-to-End System Architecture

C4 Container Architecture: SAP Fiori Horizon UI5 Client • nginx → Express/Spring Boot • Dual AI (Ollama Llama/Mistral) • Qdrant Vector RAG • MongoDB

Full Stack Tier

1. PRESENTATION & CLIENT TIER • SAP FIORI HORIZON (UI5 WEB COMPONENTS + TYPESCRIPT)

ShellBar & Navigation

View Switching & Session Profile

- ShellBarView.ts
- LanguageSelectorView.ts (i18n)
- ThemeStore (Morning / Evening)

SAP Fiori Floorplans

List Report Table & Object Page View

- ListReportView.ts (Filtering, Sort, Pagination)
- ObjectPageView.ts (BibTeX, Dual AI, RAG Chat)
- UploadDialogView.ts & VersionOverwrite

Reactive State & EventBus

Zero-dependency Pub/Sub & Observable Store

- appStore.ts (Catalog, Active Doc, Dialogs)
- core/eventBus.ts (Event Lifecycle)
- backendStore.ts (Mock vs REST mode)

BackendGateway & Adapters

Unified Contract Driver for UI5 Floorplans

- BackendGateway.ts (Typed Dispatch)
- RestBackendAdapter.ts (/api/v1/* fetch)
- MockBackendAdapter.ts (In-memory fallback)

REST / JSON / Multipart (nginx :80)

2. INGRESS & SERVICE ORCHESTRATION TIER (NGINX :80 & EXPRESS.JS :3000 & SPRING BOOT :8080)

nginx + Express Gateway Server

Single Ingress (nginx :80) & Static Host

- nginx routes /api/v1/*, /actuator/* → :8080
- server.ts (Express, Port 3000 on 0.0.0.0)
- /docs Static Hub (Javadoc & TypeDoc)
- Stream Endpoints (ZIP, BPMN, PNG, PDF)

REST API Controllers (Spring Boot :8080)

Endpoints conforming to OpenAPI 3.0

- GET/POST /api/v1/documents
- POST /api/v1/documents/:id/summarize
- POST /api/v1/chat (Conversational RAG)
- GET /api/v1/auth/me (Keycloak OIDC Profile)

Services & Extraction Engine

Parsing, Normalization & Formatting

- pdf-parse Verbatim Stream Extractor
- BibTeX Heuristic + LLM Extractor
- LaTeX BNF Key & Entry Formatter
- GUID & In-Place Version Management

Static BPMN 2.0 Export

Served/Downloaded, No Embedded Engine

- Process_DocumentIngestionRAG.bpmn
- External Camunda Modeler Compatibility
- BpmnModalView.ts (View / Download XML)
- No JavaDelegate / Zeebe Runtime Present

Inference Prompts & Embeddings

3. GENERATIVE AI & VECTOR RETRIEVAL TIER (LOCAL OLLAMA SPRING AI)

Ollama Llama 3.3 (70B Instruct)

Analytical Decomposition & Research Synthesis

- Methodological Critique, Findings & Quantitative Benchmarks
- Duration Tracking, Telemetry & Latency Recording

Ollama Mistral Large (2411)

Executive Summary & Operational Takeaways

- High-level Abstraction & Target Audience Recommendations
- Parallel Asynchronous Branch in Ingestion Pipeline

Persistence & Audit Logs

4. PERSISTENCE, VECTOR EMBEDDINGS & SECURITY IDENTITY TIER

Qdrant Vector Database

Dense Semantic Search & Cosine Retrieval

- Collection: library_embeddings (HTTP 6333 / gRPC 6334)
- Paragraph Chunking (~350 characters/chunk)
- Top-K Retrieval with Excerpt Citations

MongoDB Document Repository

Document Entities, BibTeX & Dual Summaries

- Collection: documents (GUID primary key)
- Version lineage: previousVersionGuid & versionNumber
- MongoTemplate Dynamic Queries & Multi-Field Search

Keycloak OIDC & File Storage

OAuth2 Resource Server & Asset Management

- Realm: personal-library-realm (JWT Decoding)
- Roles: LIBRARY_ADMIN, CHIEF_RESEARCHER
- Raw PDF/MD/DOCX Local / S3 Blob Storage